

Fake it til’ you Classify it: Diffusion Generated Images for Breast Cancer Histopathology

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1. Introduction

Deep learning excels across many domains, but scarce-data settings hold back many of its applications, nowhere more than in histopathology. Labeling tissue slides requires trained pathologists, making annotation slow and costly, and because the images come from patient biopsies, privacy and data-sharing rules limit how freely datasets move between institutions. Many classification tasks must therefore be learned from only a few hundred labeled images, much of which cannot leave the institution that collected it.

The usual response is data augmentation, but conventional transforms, flipping, rotation, and color jitter, only alter images that already exist. They make a dataset look more diverse without adding genuinely new visual information or expanding the underlying distribution the model learns from.

Diffusion models offer a way past this ceiling, generating realistic class-conditioned images [5, 11] that could either supplement a small dataset or, more consequentially, stand in for sensitive records: a generator trained on private slides can produce images of no real patient, letting institutions share synthetic data or the generator itself. Both uses rest on one question: can synthetic patches carry enough diagnostically meaningful signal to substitute for real data, or is their realism only surface-deep, adding noise that hurts a downstream classifier? This is not obvious: a patch can look convincing while losing the subtle features that separate, for instance, in-situ from invasive carcinoma.

We study this directly. Holding the total training-set size fixed at each of several budgets, we compare a Vision Transformer trained on real patches alone against one in which a fixed quantity of real patches is replaced by synthetic ones, for breast cancer histology. Fixing the total size isolates data source from data quantity. We ask: when, if ever, does substituting synthetic patches for real ones preserve downstream performance, and how does that depend on how much real data is available?

Our contributions are:

- A controlled, size-matched comparison of all-real against synthetic-augmented training across four real-data budgets, holding total size fixed within each budget so that data source is the only variable.
- An evaluation of whether per-class generation quality (FID) predicts per-class downstream behavior.
- An analysis of how synthetic data reshapes the classifier’s learned representations, examining *why* performance changes, not just whether it does.

2. Related Work

2.1. BACH: Breast Cancer Histology Challenge [1]

This paper defines the BACH benchmark we use as our testbed: 400 H&E-stained microscopy images across four classes (Normal, Benign, In Situ Carcinoma, Invasive Carcinoma), plus ten annotated whole-slide images. Most of it documents what the top challenge teams did, and the pattern is consistent—pre-trained CNNs (VGG, Inception, ResNet, DenseNet) fine-tuned from ImageNet, with data augmentation, color normalization, and ensembling. Key takeaways: transfer learning dominates in low-data medical imaging, large-context inputs beat small patches, ensembles help broadly, and the best systems match expert pathologists. They also flag that many teams’ validation accuracies ran $\sim 10\%$ optimistic versus the held-out test set due to information leakage across splits.

This matters because it defines our dataset and task. 400 images is exactly the low-data regime that motivates synthetic augmentation: with so few examples per class, supervised training already leans hard on transfer learning and augmentation, leaving room for diffusion-generated samples to add information that flips and crops cannot. Their top ensembled CNN baselines ($\sim 87\%$ accuracy) give us a concrete number to measure against.

2.2. Synthetic Data from Diffusion Models [2]

Azizi et al. fine-tune Imagen on ImageNet into a class-conditional generator, then train classifiers on its synthetic

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samples. Three results: (1) fine-tuning the pre-trained model beats training from scratch, hitting state-of-the-art FID (1.76) and Inception Score (239) at 256×256 ; (2) classifiers trained on only synthetic data reach a new state-of-the-art Classification Accuracy Score (64.96 at 256×256 , 69.24 at 1024×1024), closing much of the gap to the real-data ceiling near 73; (3) mixing synthetic with real boosts ResNet/ViT top-1 by 0.3–1.8%, but the benefit peaks around a $2\times$ synthetic-to-real ratio and reverses past $4\text{--}5\times$. Two sampling choices matter for downstream utility: classifier-free guidance should be far lower than for nice-looking images ($\sim 1.0\text{--}1.25$, not 8–10), and log-variance mixing aids diversity.

This is the closest existence proof for what we are attempting on BACH, and most of our design follows from it. The core argument is that adapting a big pre-trained diffusion model beats training one from scratch on the target domain—this is why we use LoRA on Stable Diffusion rather than training a latent diffusion model on 400 BACH patches directly; the data-scale gap is even more extreme in our setting. We borrow their methodology: sweeping guidance in the 1.0–3.0 range, reporting both FID and downstream accuracy, and setting up our mixing matrix to look for the same inverted-U pattern. The main difference is headroom: ImageNet classifiers are near-saturated, so their absolute gains are modest ($\sim 1.78\%$ on ResNet-50), whereas BACH’s 400 images leave far more room.

2.3. Machine Learning for Synthetic Data Generation [8]

Lu et al. survey the field around motivations (scarcity, quality, privacy), application domains, generative model families, privacy risks and defenses, fairness issues, and evaluation methods. The closing section covers open challenges: evaluation metrics, outlier coverage, bias propagation, and memorization—where models regenerate training images almost verbatim.

This serves as our framing reference in three places. The motivations section establishes low-data synthetic augmentation as a recognized response to scarcity, justifying that applying it to BACH is a known problem. The evaluation chapter names the strategies we combine: our core design is Train-on-Synthetic-Test-on-Real, while our FID-vs.-accuracy comparison adds the statistical-distance angle, and the review argues combining strategies gives a fuller picture than either alone. Finally, it flags that Stable Diffusion can memorize and regenerate training images, which in our utility-focused setup would mean apparent gains could partly reflect retrieval of near-duplicates from SD’s web-scraped pretraining set—this is why we added a nearest-neighbor sanity check between generated samples and the training set.

3. Methods

3.1. Dataset

We evaluate on the BACH (Breast Cancer Histology) dataset from the ICIAR 2018 Grand Challenge [1]. It contains 400 labeled breast-tissue microscopy images evenly distributed across four diagnostic classes: Normal, Benign, In Situ Carcinoma, and Invasive Carcinoma (100 images per class). We split these 400 into training, validation, and test sets described in Sec. 3.3. Each image is 2048×1536 pixels, so much of each slide consists of cellular background or regions uninformative for downstream diagnosis. Classifying the four classes is clinically meaningful, and differentiating between them can be very subtle—separating In Situ from Invasive Carcinoma, for instance, can challenge even trained pathologists. The small dataset size also makes it a realistic low-resource setting where data augmentation matters most.

3.2. Patch Extraction and Tissue Filtering

Whole BACH slides are too large to feed directly into a Vision Transformer, and downscaling an entire slide to the model’s input resolution would destroy the cellular detail crucial to diagnosis. We instead tile each image into 512×512 -pixel patches using a sliding window with 50% overlap (stride 256), yielding roughly 35 patches per image. This patch size preserves fine cellular detail while avoiding upscaling artifacts, and mirrors how pathologists examine slides for abnormalities [1].

A large portion of each slide consists of cellular background, which carries no information for breast cancer diagnosis. We score each candidate patch by its tissue-to-background ratio and discard background-dominated patches. Tissue ratio is computed by applying Otsu’s threshold [9] to the saturation channel of the HSV color space: H&E-stained tissue (hematoxylin and eosin) appears highly saturated while the background appears near-white with low saturation, so the saturation channel separates the two without manual annotation. Patches with tissue ratios below 0.40 are discarded; from the remaining pool we keep the top 12 patches per image by tissue fraction.

3.3. Data Splitting and Class Balancing

We split the 400 images into training, validation, and test sets at a 70/10/20 ratio per class, yielding 70 training, 10 validation, and 20 test images per class. The split is performed at the image level before patch extraction to prevent leakage between patches drawn from the same image. The test set of 80 images (20 per class) is held out until all experimental conditions are finalized.

Per-class patch counts varied from $\sim 360\text{--}590$ because each image’s tissue content differs, leaving classes imbalanced in patch count. To prevent this from confounding

comparisons, we cap all classes at the smallest available patch count and sample that many patches uniformly at random from each class. For the baseline, this yields 360 patches per class (1440 total). Validation and test sets use all available patches without capping, since image-level evaluation is the primary metric and the image-level split is already balanced.

3.4. Experimental Design

The central question is whether diffusion-generated synthetic patches are as informative as real patches when used as a data supplement, and whether the answer changes with the proportion of synthetic data. We therefore train classifiers at four real-data budgets, 50, 100, 150, and 250 patches per class, each under two conditions: a *real-supplement baseline*, which adds 100 real patches per class on top of the budget, and a *synthetic-supplement condition*, which replaces those 100 real patches with 100 diffusion-generated patches. This yields 8 total training conditions, summarized in Table 1.

Because both conditions add exactly 100 patches per class at every budget, the total training-set size is identical within each budget. Any difference in performance is therefore attributable solely to data source. Sweeping the real budget from 50 to 250 patches per class traces how the relative value of synthetic data changes with data scarcity: at 50 patches per class the supplement doubles the available training data, whereas at 250 it represents a smaller relative addition.

The synthetic patches are generated by a diffusion model fine-tuned on the training-split patches, described in Sec. 3.7. Validation and test sets remain 100% real in every condition, so evaluation never depends on synthetic data. All eight classifiers are initialized from the same ImageNet-pretrained checkpoint and trained with the same architecture and procedure.

3.5. Classifier Architecture

We use Google’s ViT-B/16 [3], a Vision Transformer pre-trained on ImageNet-21k, with 16×16 -pixel patches and 224×224 input resolution. We choose a transformer over a convolutional backbone because histopathology classification benefits from global feature and structure detection, which self-attention captures more naturally than the spatial locality bias of a CNN. We replace the original ImageNet head with a randomly initialized 4-class linear head for classification.

We fine-tune the ViT by applying Low-Rank Adaptation (LoRA) [6] to the Q, K, V projection matrices of every attention layer, freezing all other backbone weights. With rank $r = 8$, scaling factor $\alpha = 16$, and dropout 0.10, this reduces trainable parameters from 86M to 2.4M, keeping overfitting in check on our small training set.

3.6. Preprocessing and Training Procedure

3.6.1. Image Preprocessing and Augmentation

Patches are resized from 512×512 to the model’s native 224×224 and normalized to match the mean and standard deviation used during pretraining. Augmentations are applied on-the-fly at training time: random horizontal and vertical flips, random rotation over the full 360° range, and mild brightness and contrast jitter. Hue shifts are excluded since they would corrupt the H&E stain-color information the classifier relies on. The augmentation policy is identical across all training conditions.

3.6.2. Evaluation Metrics

We assess predictions at the image level, following the BACH benchmark. Since each slide is represented by multiple patches, we aggregate patch predictions back to the source image before scoring. For a given slide with n patches, we average the per-patch softmax distributions $p_1, \dots, p_n$ into a mean distribution $\bar{p}$ and predict whichever class has the highest mean probability. This mirrors how clinical histopathology pipelines reason over a whole slide and yields cleaner results by averaging out individual patch errors.

3.6.3. Training Settings

We train with AdamW (learning rate 1×10^{-4} , weight decay 0.01) for up to 40 epochs with a batch size of 32. We use early stopping on the validation image-level macro-F1 with a patience of 8 epochs, saving the checkpoint with the highest value. We monitor image-level rather than patch-level F1 specifically because patch-level F1 tends to keep climbing as the model overfits individual patches while image-level generalization degrades.

3.7. Synthetic Patch Generation

To produce the synthetic patches used in the mixed conditions of Sec. 3.4, we fine-tune a text-to-image latent diffusion model to generate class-conditional histology patches. Training a diffusion model from scratch is infeasible at this data scale (~ 340 training slides), so we instead adapt a pre-trained Stable Diffusion 1.5 [11] through Low-Rank Adaptation, leveraging its strong natural-image prior and steering generation toward each diagnostic class with a per-class text prompt. This mirrors the transfer-learning strategy we use for the classifier (Sec. 3.5): the bulk of the model stays frozen, and a small adapter learns the histology-specific shift in distribution.

3.7.1. Patch Preparation and Conditioning

The generator is trained on the same training-split patches as the classifier, extracted with the tissue-filtering procedure of Sec. 3.2. Patches are normalized to $[-1, 1]$. Because histopathology has no canonical orientation, we apply

Table 1. Training conditions. At each real budget, the baseline supplements with 100 real patches per class and the augmented condition substitutes 100 synthetic patches. Total training size is held equal within each budget.

Real budget (patches/class)	Condition	Real patches/class	Synthetic patches/class	Total patches/class
50	Baseline (+real)	150	0	150
50	Augmented (+synthetic)	50	100	150
100	Baseline (+real)	200	0	200
100	Augmented (+synthetic)	100	100	200
150	Baseline (+real)	250	0	250
150	Augmented (+synthetic)	150	100	250
250	Baseline (+real)	350	0	350
250	Augmented (+synthetic)	250	100	350

horizontal and vertical flips combined with 90° rotations as label-preserving augmentation, together with mild color jitter to capture stain variability. We balance the classes down to the minority patch count during fine-tuning so the generator does not over-represent the more patch-rich classes.

Class conditioning is supplied entirely through text. Each diagnostic class is mapped to a fixed prompt of the form “*histopathology H and E stained microscopy image of [class] of the breast.*” Although the four prompts share most of their tokens and therefore have a high global embedding similarity (≈ 0.9 cosine), cross-attention operates per token position, and the cosine restricted to the differing token positions is far lower (≈ 0.45). We therefore keep the text encoder frozen: the pretrained encoder already provides a usable per-class signal, and only the UNet adapter needs to learn the mapping from these tokens to the corresponding tissue morphology.

3.7.2. LoRA Fine-Tuning

We freeze the VAE, the text encoder, and the base UNet weights, and train rank-16 LoRA adapters on the UNet’s Q, K, V attention projection matrices, amounting to roughly 3M trainable parameters. Training uses the standard DDPM objective [5]: a patch is encoded into the VAE latent space, noised at a randomly sampled timestep, and the UNet predicts the added noise under a mean-squared-error loss. We train with early stopping (patience 5) on a held-out validation denoising loss, with noise and timesteps seeded per batch so the metric reflects genuine model improvement rather than sampling variance.

3.7.3. Sampling

To generate, we reattach the trained LoRA adapters to the base model and sample with a DPMSolver scheduler [7] for 30 denoising steps at a classifier-free guidance scale of 4.0. We deliberately keep guidance low: high guidance on histology sharpens class-typicality at the expense of sample diversity, and diversity is precisely the property we want from an augmentation set. Generated images are written into a class-folded directory that maps directly onto the classi-

fier’s dataset format. For the experiments reported here we generated 100 patches per class (400 total).

3.7.4. Validating Realism and Memorization

Because the synthetic patches are derived from real patient slides, we validate the generator along two orthogonal axes: how *realistic* the samples are, and whether they *memorize* training data. Both are measured relative to held-out real data rather than against absolute thresholds, since absolute scores are not meaningful in this regime.

Realism — Fréchet Inception Distance. We report clean-FID [10] between the synthetic set and both the training and validation patches, per class and pooled (Table 2). The absolute values are high (~ 140 – 240), which is expected: FID’s Inception backbone [4] is trained on ImageNet natural images and is far out of domain on histology, and FID is biased upward at the ~ 100 samples/class available here. The values are therefore informative only in relative terms. Across classes, Benign is the most realistic and the carcinoma classes (In Situ, Invasive) the least, consistent with their more varied and complex morphology. More importantly, the distance to held-out validation patches tracks the distance to training patches closely (150.5 vs. 144.0 pooled); had the generator memorized the training distribution, FID-vs.-train would be substantially lower than FID-vs.-val.

Table 2. Clean-FID of the synthetic set against real training and held-out validation patches. Lower is closer to real. Read relatively, not as absolute quality scores.

Class	FID vs. train	FID vs. val
Normal	188.96	210.05
Benign	166.06	197.76
In Situ	204.47	253.35
Invasive	237.01	228.85
Overall (pooled)	143.97	150.53

Memorization — Nearest-Neighbor Analysis. To rule out reproduction of individual patches, we run a two-stage nearest-neighbor search: InceptionV3 2048-d features with a cosine kNN to retrieve the top-5 training candidates for each synthetic patch, followed by an LPIPS [12] re-ranking to identify the perceptually closest real patch. We compare the median synth→train LPIPS distance against a real↔real baseline and flag any synthetic image closer than the 1st percentile of that baseline. Across all classes the synth→train distances sit within the real↔real band (Table 3): the single closest synthetic image had an LPIPS of 0.276, far from the ~ 0 of a pixel-level copy. The only flags were 6 Benign patches at LPIPS ≈ 0.31 , indicating mild mode-convergence toward one prototypical Benign appearance rather than reproduction of specific patient data.

Table 3. Synthetic-to-training nearest-neighbor LPIPS versus the real-to-real baseline. Distances within the same band indicate generalization; a memorized copy would show synth→train near 0 while the baseline stayed ~ 0.4 .

Class	synth→train	real↔real	flagged
Normal	0.416	0.437	0/100
Benign	0.397	0.434	6/100
In Situ	0.405	0.435	0/100
Invasive	0.343	0.340	0/100
Overall	0.392	0.417	6/400

Taken together, these checks show that the generator produces samples that are plausibly on-distribution yet demonstrably generalizing rather than memorizing—the two properties required for synthetic patches to serve as a sound augmentation source.

4. Experiments

We evaluate all eight conditions from Sec. 3.4 on the held-out test set of 80 images (20 per class). Across every condition the classifier is trained and evaluated following Sec. 3.5–3.6, with image-level macro-F1 as the primary metric. The experiments are organized around three questions: whether synthetic data is a viable substitute for real data at any budget (Sec. 4.1), which diagnostic classes are most and least affected by the substitution (Sec. 4.2), and whether per-class FID predicts downstream utility (Sec. 4.3).

4.1. Real vs. Synthetic Supplement

For each real-data budget (50, 100, 150, and 250 patches per class), we compare the all-real baseline against the synthetic-augmented condition. Both conditions train on the same total number of patches per class, differing only in whether the 100-patch supplement is drawn from real or synthetic data.

The key interest is the performance gap between the two conditions at each budget and how it evolves as the real-data budget grows. A widening gap at higher budgets would indicate that synthetic data becomes a progressively worse substitute as more genuine training examples are available. A stable gap would suggest the synthetic patches carry a comparable benefit regardless of real-data availability.

4.2. Per-Class Analysis

Aggregate macro-F1 can hide where the real-versus-synthetic gap actually comes from. We break down macro-F1 into per-class F1 scores and compare them between the baseline and synthetic-augmented conditions at each budget.

BACH’s four classes carry very different levels of visual complexity: Normal and Benign tissue exhibit relatively regular architecture, while In Situ and Invasive Carcinoma encompass heterogeneous growth patterns and cellular arrangements that vary substantially even within a class. If the harm from synthetic substitution concentrates on In Situ and Invasive, it would suggest the diffusion model is better at capturing simpler, more predictable tissue morphology than the kind of variability that makes carcinoma classification hard.

4.3. Generation Quality as a Predictor of Downstream Utility

Sec. 3.7.4 established a per-class ranking of generation quality via FID: Benign patches were generated most faithfully (FID 166.06 vs. train), followed by Normal (188.96), In Situ (204.47), and Invasive (237.01). We examine whether this ordering predicts downstream behavior—specifically, whether classes with lower FID also show a smaller F1 gap between the all-real and synthetic-augmented conditions.

A consistent correlation would suggest that FID is a useful signal for augmentation utility. Its absence would indicate that distributional realism as measured by FID does not fully capture what makes a synthetic patch useful for downstream classification, and that factors like intra-class morphological variability are a better predictor.

5. Results

5.1. Interpreting the Results

At the smallest budget the two arms were essentially tied: 0.519 image-level macro-F1 for all-real vs. 0.516 for synthetic-augmented at 150 patches per class (Fig. 1). The all-real advantage widened as the budget grew: 0.053 at 200, 0.133 at 250, and 0.163 at 350 (0.627 vs. 0.464). Both arms dipped at 250 before all-real recovered to its best score, so the curve is non-monotonic, but the trend is consistent: substituting synthetic for real patches costs almost

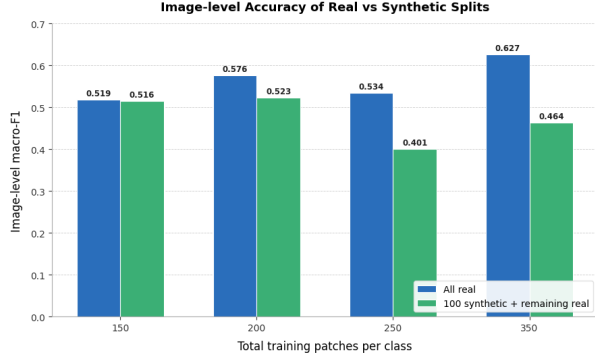


Figure 1. Image-level macro-F1 of the all-real condition (blue) versus the synthetic-augmented condition (green; 100 synthetic patches per class plus the remaining real patches) at four matched total training-set sizes (150, 200, 250, and 350 patches per class). Within each budget the total training-set size is held constant, so the only difference is data source. The all-real arm matches or exceeds the synthetic-augmented arm at every budget, and the gap widens as the real-data budget grows.

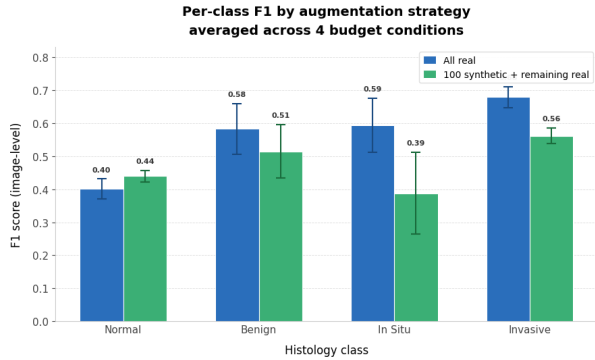


Figure 2. Per-class image-level F1 by augmentation strategy, averaged across the four budget conditions, for the all-real (blue) and synthetic-augmented (green) conditions. Error bars denote variation across budgets. Synthetic augmentation improves only the Normal class and harms the three remaining classes, with the largest and least stable drop on In Situ Carcinoma.

nothing when real data is scarcest and increasingly hurts as more real data becomes available, never matching the all-real condition beyond the smallest budget.

The per-class breakdown (Fig. 2) shows why. Averaged over the four budgets, only Normal improved (0.40 $\rightarrow$ 0.44); the carcinoma classes were harmed most: Benign -0.07 , Invasive -0.12 , and In Situ most of all (0.59 $\rightarrow$ 0.39), with by far the widest spread across budgets. These outcomes do not track per-class FID (Table 2): the most realistic class by FID, Benign, was still hurt, and the only beneficiary, Normal, is merely middling on FID. A weak ordering appears among the carcinomas against validation FID (In Situ worst on both axes), but it breaks under train

FID, so with only four classes and visible budget-to-budget variance we read FID as an unreliable predictor of downstream effect. The more consistent factor is morphological complexity: the homogeneous, low-ambiguity Normal class gains from added variation, whereas the diagnostically subtle carcinoma classes are where substitution does the most and least predictable damage.

Together these findings point away from surface realism, which FID measures through an out-of-domain ImageNet backbone, and toward a representational mechanism: the generator collapses diagnostically meaningful intra-class variation toward a class prototype, and the classifier inherits these smoothed features. Synthetic patches can therefore look realistic while losing exactly the fine, subtype-distinguishing morphology the carcinoma classes depend on.

5.2. Limitations

Three factors constrain the result. First, the dataset is small for fine-tuning a ViT. The BACH dataset does not label the official challenge test set, so we carved our own train/val/test split from the 400 labeled images, further reducing available training data. This increases variance in all conditions, though it also more faithfully represents the low-resource scenario we are studying.

Second, histopathology is far out of domain from the ImageNet pretraining that both the classifier and the Stable Diffusion prior depend on, inflating absolute FID values. Third, BACH images of similar subtypes carry adjacent file indices, so the slide-level split may have placed more diverse tissue representations in the test set than the training set, biasing evaluation against any model that learned only the training distribution.

5.3. Discussion

The central finding is a negative one: under a size-matched substitution, diffusion-generated patches did not stand in for real patches and actively degraded the downstream classifier, with the harm growing as more real data became available. At least on BACH at this scale, a generator trained on private slides could not produce a synthetic corpus that substitutes for real records without a measurable diagnostic cost. We believe the most likely cause is intra-class mode convergence: the generator captured each class’s recurring, prototypical appearance while smoothing over the subtle subtype-level variation that distinguishes, e.g., In Situ from Invasive Carcinoma—the same averaging visible in the mild Benign mode-convergence flagged in Sec. 3.7.4. Promising directions include subtype-aware conditioning, larger or multi-institution training corpora, and using synthetic data as a supplement to a full real set rather than a replacement for it.

6. Future Work

Our results point to a single limiting factor: the generator preserves surface realism but not the fine, class-distinguishing morphology that classification depends on, failing most on the subtle carcinoma classes. We trace this to three specific limitations, each pointing to a direction for improvement.

Coarse labeling. BACH provides one label per image across four broad classes, but each class spans several unannotated morphological subtypes. Seeing only the coarse label, the text-conditioned generator cannot keep these subtypes apart and collapses toward an averaged, prototypical appearance per class—the mode-convergence visible in the Benign nearest-neighbor flags (Sec. 3.7.4). Annotating finer labels, such as tumor grade or histological subtype, and conditioning on them would let the model represent the within-class variation it currently discards, exactly what the carcinoma classes need.

Patch selection. We select patches by tissue-to-background ratio (Sec. 3.2), which favors tissue density over diagnostic content, so a dense patch of unremarkable stroma is kept over a sparser one containing the discriminative tumor region. Both generator and classifier therefore train on a pool only loosely aligned with class-separating features. Selecting by diagnostic relevance, via pathologist-annotated regions, attention maps, or a learned saliency model, could concentrate the training signal on the regions that matter.

Shared generator. A single Stable Diffusion model served all four classes, steered by near-identical prompts (Sec. 3.7.1), forcing one set of adapters to model every class at once and inviting cross-class interference that blurs class-specific structure. A separate generator per class or per subtype, combined with finer labels, could devote full capacity to one distribution and sharpen the morphology distinguishing, for example, In Situ from Invasive Carcinoma, at the cost of more training and less data per model.

These limitations share a root cause: each pushed the generator toward the average rather than the diagnostically distinctive, which is why synthetic patches looked realistic yet failed downstream. Addressing them would move synthetic generation from merely realistic to diagnostically relevant—the precondition for synthetic data to supplement, and eventually substitute for, real data without the downstream cost we observed.

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